^{35}Cl(n,n′γ) 1989Ge09,1984El12,1966Ni03

1984El12,1989Ge09: fast neutrons of diameter 26 mm were produced from the IRT-2000 reactor at the Institute for Nuclear Research and Nuclear Power of the Bulgarian Academy of Sciences in Sofia. Two ^{35}Cl targets were placed at 45° and 135° with a distance of 70 cm apart. γ rays were detected using a Ge(Li) detector. Measured Eγ. Deduced lifetimes for 4 levels using the Doppler Shift Attenuation Method (DSAM) and transition strengths.

1966Ni03: 2.5-4.2-MeV neutrons were produced via the T(p,n) reactions with protons from the university of Kentucky 6-MeV accelerator. Targets were a cylinder of CCl_4 and a 5.6-cm-diam sphere of C_2Cl_6. γ rays were detected using a 6.35-cm by 6.35-cm NaI(Tl) detector and 7.5-cc Ge(Li) detector. Measured Eγ, Iγ, γ(θ). Deduced levels, J.

1971Fr05: neutrons up to 10 MeV were produced from a 5-Ci Pu-Be neutron source and bombarded table salt targets. γ rays were detected using a Ge(Li) detector. Measured 2 Eγ and deduced 2 levels.

Others: 1955To32, 1977Ko19, 1983Gl07, 1992Be60 (cross sections).

^{35}Cl Levels

E(level)†	Jπ#	$T_{1/2}$‡
0	$3/2^+$	
1219	1/2	139 fs *56*
1762	5/2	
2646	7/2	187 fs *62*
2694		43 fs *6*
3002		50 fs *9*
3163		

† From a least-squares fit to γ-ray energies.
‡ From 1984El12 and 1989Ge09 using DSAM.
# From γ(θ) in 1966Ni03, except for the ground state from the Adopted Levels.

γ(^{35}Cl)

Eγ	E_i(level)	J_i^π	E_f	J_f^π	Comments
1219	1219	1/2	0	$3/2^+$	
1762	1762	5/2	0	$3/2^+$	Eγ: from 1966Ni03. A_0=+0.942 *8*, A_2=−0.148 *23* (1966Ni03).
2646	2646	7/2	0	$3/2^+$	A_0=+1.078 *5*, A_2=+0.165 *12* (1966Ni03).
2694	2694		0	$3/2^+$	
3002	3002		0	$3/2^+$	
3163	3163		0	$3/2^+$	Eγ: from 1966Ni03.

^{35}Cl(n,n′γ) 1989Ge09,1984El12,1966Ni03

Level Scheme